MIT-ADT UNIVERSITY
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MIT SCHOOL OF COMPUTING

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

AI-Based Student Performance Monitoring & Adaptive Test Platform for Placement Preparation

Shourya Shah, Yash Shah

Faculty Guide:

Dr. Ganesh Pathak

User

What User Says

- I don't know which topic I am weak in.
- All tests feel the same, not helping me improve.
- I only see marks, not real feedback.
- I don't know if I am placement ready.

What User Thinks

- I keep repeating the same mistakes.
- I need practice based on my weak a
- These platforms don't understand my level.

Am I improving or just guessing?

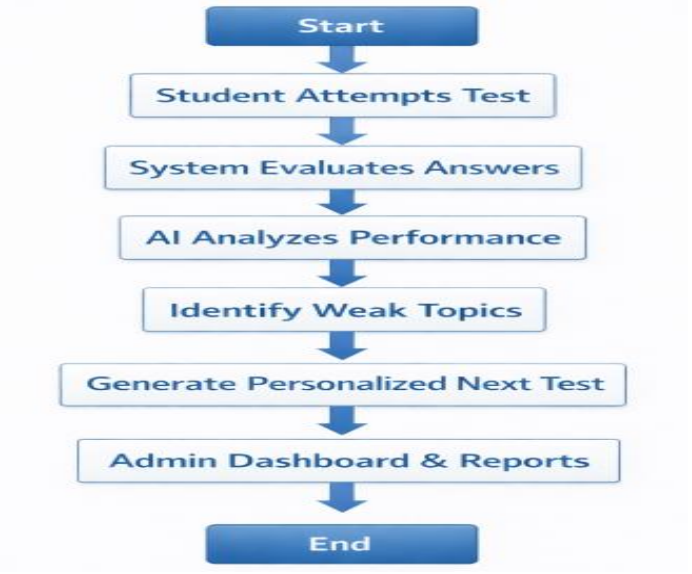
What User Does

- Attempts random mock tests
- Skips difficult questions
- Revises without clear direction
- Depends on external resources

What User Feels

- Confused about preparation path
- Frustrated due to low scores
- Anxious about placements
- Lacks confidence in subjects

Sr. No.	Requirement	Proposed Solution
1	Personalized Student Evaluation	We will build an AI module that analyzes each student's test attempts in real time. It will identify topics, repeated mistakes, and time-consuming the actual learning gaps of every individual.
2	Adaptive Test Generation	The system will automatically generate the next test performance. More previous questions will be selected from weak areas with a mix of revision to create a personalized learning path.
3	Admin & Placement Analytics	We will provide an admin dashboard where placement officers can monitor student progress, compare divisions, and identify weak subjects across batch for data-driven training decisions.
4	AI-Based Readiness Prediction	AI models will classify students as strong, average, or weak or predict placement readiness. The systent focus areas and also suggest focus areas improvement plans for better outcomes.



Problem statement

Current placement preparation platforms treat all students equally and provide static question sets without any personalization. They lack analysis of wrong and unattempted questions, identification of topic-wise weaknesses, and adaptive tests based on individual performance. There is also no centralized monitoring system for placement officers and no intelligent insights beyond simple marks. This results in poor personalized learning and difficulty in identifying placement readiness at both student and batch levels.

Proposed Solution

We propose an AI-driven adaptive MCQ platform for placement subjects such as DBMS, OS, CN, DSA, Aptitude, Logical, and Verbal, which will track student attempt patterns and analyze mistakes and unattempted questions. The system will dynamically generate the next test based on individual performance, provide admin dashboards for batch analytics, and predict placement readiness using AI models, creating a personalized learning path instead of a one-size-fits-all approach.

Scope and Feasibility

The proposed system focuses on individual student monitoring, adaptive test generation, and batch analytics across DBMS, OS, CN, DSA, and Aptitude. It is feasible as a web-based platform using AI/ML to analyze responses, a scalable question bank, and role-based access for students and administrators. The system can be integrated into placement workflows to provide personalized learning and data-driven decisions.